EXERCISE SHEET 4

GROUP AND RING THEORY

Main questions - to be done before the seminar

4.1. Recall that a group G is *solvable* if there are subgroups $1 = G_0 < G_1 < \cdots < G_n = G$ such that G_{i-1} is normal in G_i, and G_i/G_{i-1} is an abelian group, for $i \in \{1, 2, \ldots, n\}$.

Let G be a group of order p^2q, where p and q are distinct primes. Show that G contains a normal Sylow subgroup and that G is solvable.

4.2. Let G be a simple non-abelian group and let n_p be the number of Sylow p-subgroups, where p is a prime dividing $|G|$. Show that $|G|$ divides $(n_p)!$. Using this or otherwise, prove that groups of order 48 cannot be simple.

4.3. Let P be a Sylow p-subgroup of a finite group G. If M is a normal p-subgroup, show that $M \subseteq P$. Show also that G contains a normal p-subgroup H that contains all normal p-subgroups of G.

Extra questions

4.4. Suppose P is a Sylow p-subgroup of a group G. Let $H \leq G$ be such that $|H| = p^m$, for $m > 0$. Show that $H \cap N_G(P) = H \cap P$.

4.5. Suppose P is a Sylow p-subgroup of a group G. Show that the number n_p of Sylow p-subgroups of G equals $|G : N_G(P)|$.